Global impact of rainfall on mortality: drought channel

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Purpose

Goal: estimate the impact of rainfall on mortality, globally.

- **Challenge:** the impacts of rainfall on mortality are highly heterogeneous.
- Droughts kill — but so does extreme rainfall.
- Different functions of rainfall impact mortality through different mechanisms, at different temporal scales, in different places.

A single pooled response function may average effects toward zero.

Plan

Step 1	**Paradigms** Build a series of mechanism-specific paradigms, informed by the literature, showing empirically how different mechanisms operate in different parts of the sample.
Step 2	**Model selection** Use machine learning to allow for rich heterogeneity everywhere, informed by the paradigms from Stage 1.

Current status: paradigm 1 — extreme rainfall in cities

- Bearpark et al. (2025, *Nature*): extreme daily rainfall causes large mortality impacts in Mumbai.
- Suggestive evidence of the same pattern in our sample, when restricting to admin units that resemble Mumbai (dense, urban, flood-exposed).

Paradigm 1: Cities may carry large negative effects of *extreme daily* rainfall.

Paradigm 2: drought

How should we measure drought,
and where should it matter for mortality?

Literature

The Economics of Drought

Core insight: the rainfall deficit is only the beginning of the story. Whether it becomes an actual shortage depends on storage, groundwater, irrigation, conveyance, prices, allocation rules, markets, ...

- **drought $\neq$ water scarcity $\neq$ water shortage**.
  - A drought can raise scarcity without creating a shortage that households or farms experience.
  - Conversely: high demand or poor allocation can create a shortage with no exceptional meteorological drought.

A useful frame for choosing heterogeneity dimensions and mechanisms.

[Edwards, Sanchez & Sekhri 2024]

Water Scarcity and Birth Outcomes in the Brazilian Semiarid

Rainfall deficits during pregnancy worsen birth outcomes and raise infant mortality in poor semiarid places, mainly through intestinal infections and malnutrition.

- **Key heterogeneity:** mortality effects are larger where piped-water and sanitation coverage are low (and for youngest age group).
- Measures rainfall during
- Null impacts for agriculture

[Rocha & Soares 2015]

Global WASH mapping

A data source for heterogeneity: annual 5 km estimates of water-source and sanitation categories, 90 LMICs, 2000–2017.

- Rasters plus adm1/adm2 summaries; the rasters let us aggregate to *our own* admin boundaries.
- Four categories each for water source and sanitation. Initially keep them **separate** rather than collapsing into a single WASH index — they may capture different dimensions of vulnerability.

Open question: let these measures vary through time, or take a fixed baseline / sample average?

[Local Burden of Disease WASH Collaborators 2020]

Drought and mortality in urban Brazil

Weekly mortality in 13 large Brazilian urban areas, linked to SPEI-1.

- Pooled effects small and imprecise — but stronger signals for respiratory and circulatory mortality, and among women, children and older adults.
- **Takeaway:** do not exclude cities from the drought analysis. Short dry periods may affect particular urban populations even when the average city-level relationship is weak.
- Mechanisms not identified; candidates discussed: water-system stress, dust / air pollution, dehydration, heat, socioeconomic vulnerability. A good starting list.
- **Measurement issue:** they use city-center exposure, but drought may hit a city's *source watershed or reservoir system*. How to measure that? (Proxy from James's hydrology work?)

[Salvador et al. 2022]

Defining and Measuring Drought

What is drought?

Working definition: a rainfall shortage that is *unusual* for a place and time of year, and that *persists* long enough to matter.

Three features from the literature [Wilhite & Glantz 1985] [Dracup, Lee & Paulson 1980] [WMO/GWP 2016]:

- **Relative:** 20 mm can be extremely dry in one place/month and completely normal in another. A normal dry season is not drought.
- **Persistent:** one dry day, week or even month is a rainfall shock; drought is a deficit that continues or accumulates.
- **Propagates:** precipitation deficit → low soil moisture → low streamflow / storage → crop stress → experienced water or food shortage. Related, but *not* the same exposure, nor the same time scale.

Four categories [Wilhite & Glantz 1985]: **meteorological** · agricultural · hydrological · socioeconomic. Our common cross-country exposure is **meteorological**; the others are mechanism-specific measures or consequences (agricultural drought may need a growing-season definition — to discuss).

Proposed construction: from admin-month rainfall to episodes

Dimensions worth preserving [Dracup, Lee & Paulson 1980]: onset · duration · intensity · cumulative severity · worst month ("nadir") · recovery. *Which do we need?*

1. **Admin-month rainfall.** Sum daily admin-level rainfall to monthly totals (already processed; pop-weighted as default?).
2. **Fixed reference period.** Same across the panel — but which? Projection-system "historical climate" (1981–2001? 1981–2010?). Country-specific baselines would change what the percentile means. Alternative reference periods as sensitivity checks. *Tom: rolling average* — needs precip data beyond mortality coverage years.
3. **Local rainfall percentile.** Compare each admin-month with the historical distribution for the *same admin, same calendar month* (January vs. Januaries). Solves cross-climate comparability; retain totals and mm deficit alongside.
4. **Join months into episodes.** First rule: $\leq$ 20th local percentile for ≥ 2 consecutive months (test alternatives). Per episode keep onset, duration, worst percentile, cumulative percentile and mm deficits.

Standard indices, and getting to the annual panel

Standard indices as complements?

- **SPI-3 / 6 / 12** from monthly precip [McKee et al. 1993] [WMO 2012]: widely used, comparable across climates; less intuitive than direct duration/severity. Windows map to mechanisms: SPI-3 seasonal / agriculture, SPI-12 longer-term water supply. Constructed at adm-unit level (settled).
- **SPEI** as robustness [Vicente-Serrano et al. 2010]: adds evapotranspiration — but then partly combines low precip with temperature.

Monthly episodes → annual mortality panel. Identify episodes in monthly data first, then translate to admin-year measures:

- months in qualifying drought during the year; maximum episode duration
- cumulative percentile and/or mm deficit; worst monthly percentile
- minimum SPI-3 / SPI-6 / SPI-12

Next Steps

First mechanism: household water and sanitation

The clearest place to start — we have a plausible cross-country data source.

The story: persistent low rainfall reduces water quantity or reliability → households store water or switch sources → hygiene and sanitation worsen → infection, dehydration, nutrition risks rise. Expect the strongest effects where piped-water and sanitation access are low, surface/unimproved water is common, and among infants and young children.

What we can realistically do:

- Aggregate the LBD WASH estimates to our admin units.
- Build baseline (or time-varying) shares: piped water, surface/unimproved water, sewer/septic, ...
- Interact these with drought — separately at first, not as one index.
- Focus on the youngest age group and enteric/malnutrition causes where available (only a few countries in our sample).